



Accelerated Quantum Supercomputing in Action: A Hands-On Tutorial on Scalable Hybrid Workflows

SC25 | Nov 16, 2025

What sparked your interest to attend this talk?
How familiar are you with ...?

menti.com
Code: 6311 8148



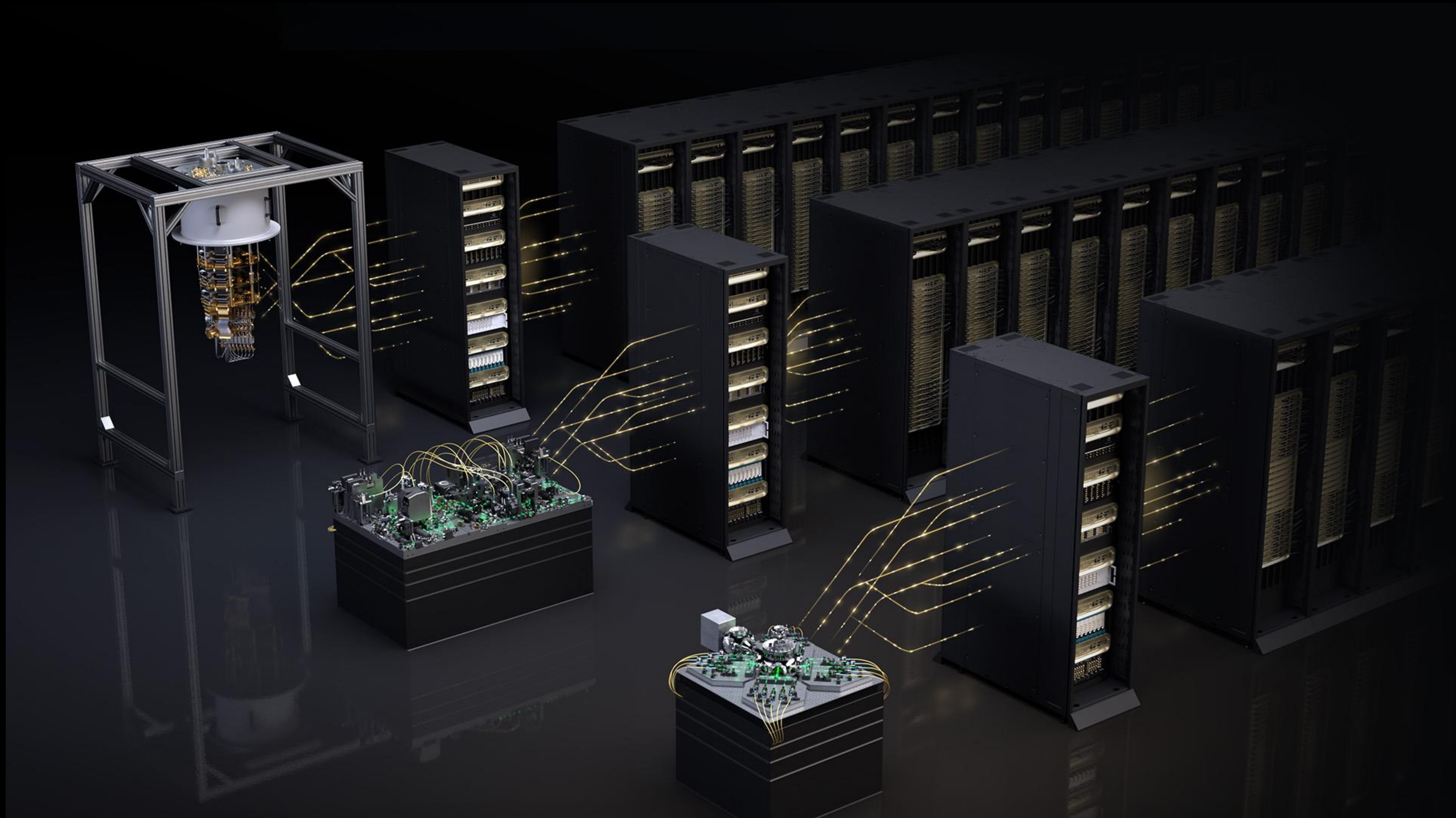
Agenda

- Hour 1: Overview of Accelerated Quantum Supercomputing and VQE

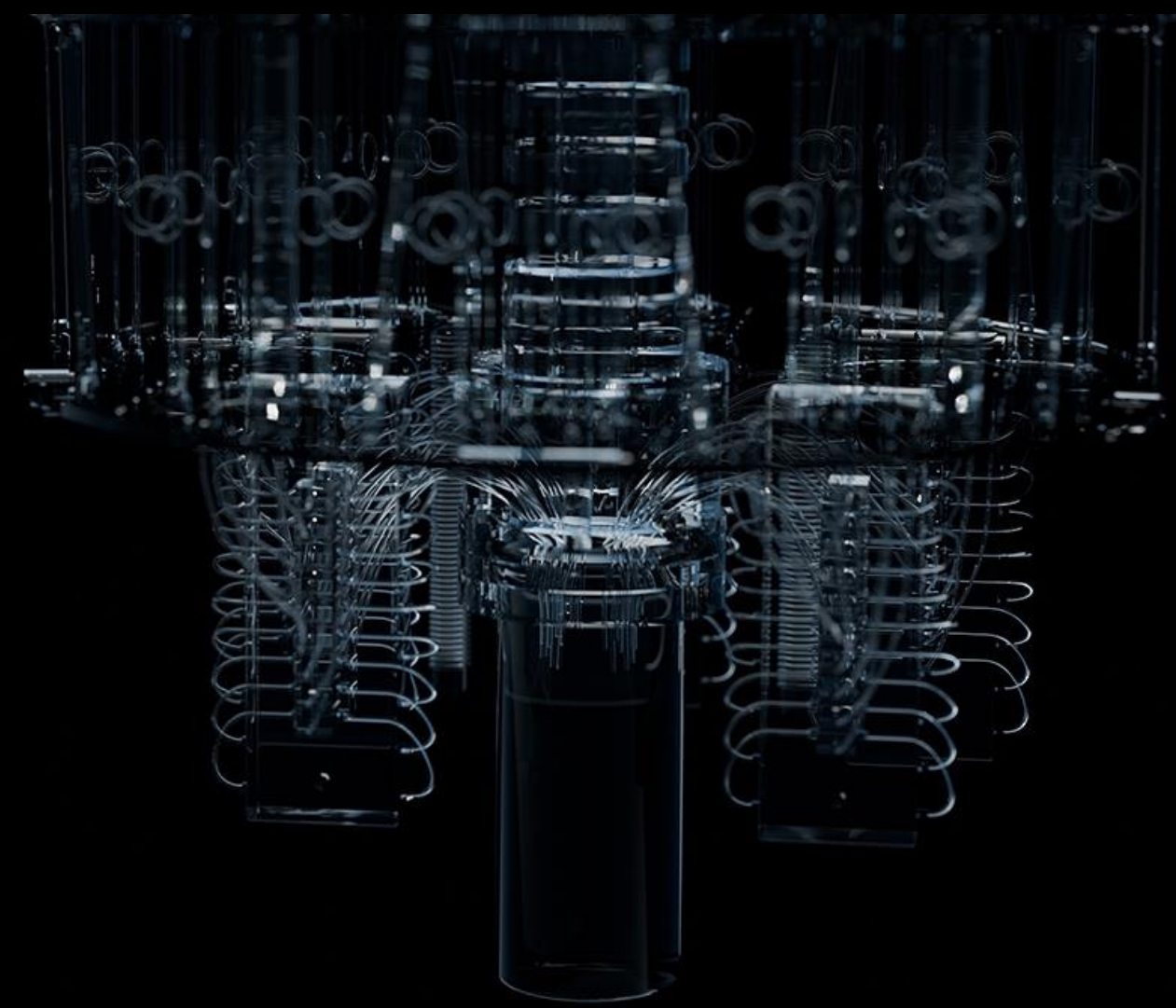
- Hour 2: GQE and AFQMC (with break between 3:00-3:30)

- Hour 3: Inflection Sqale Platform

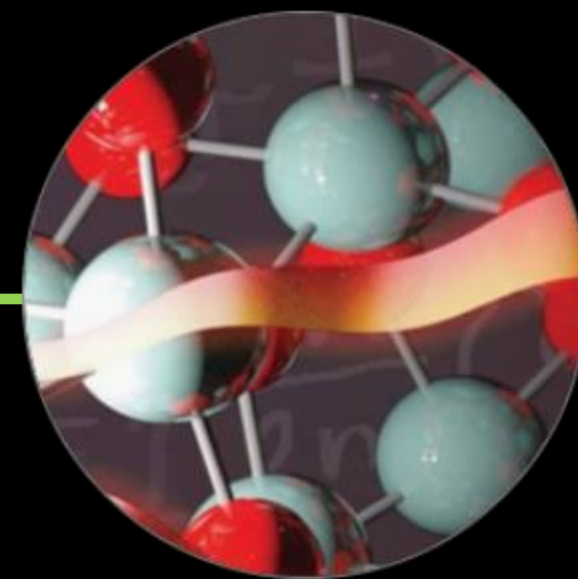
- Resources to learn more



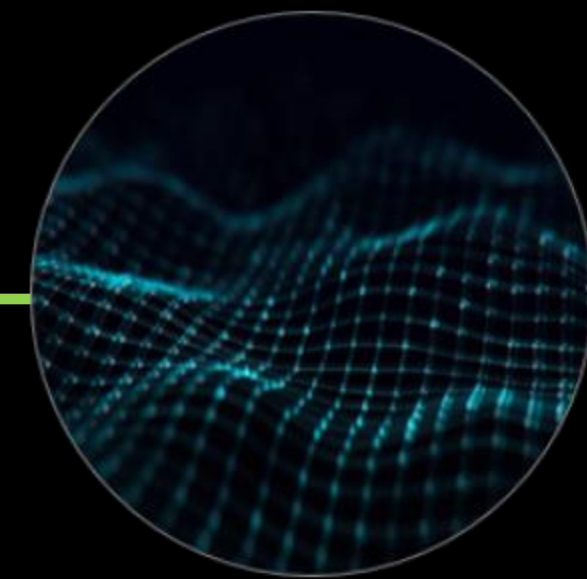
The Journey From Qubits to Supercomputers



Today's
Systems



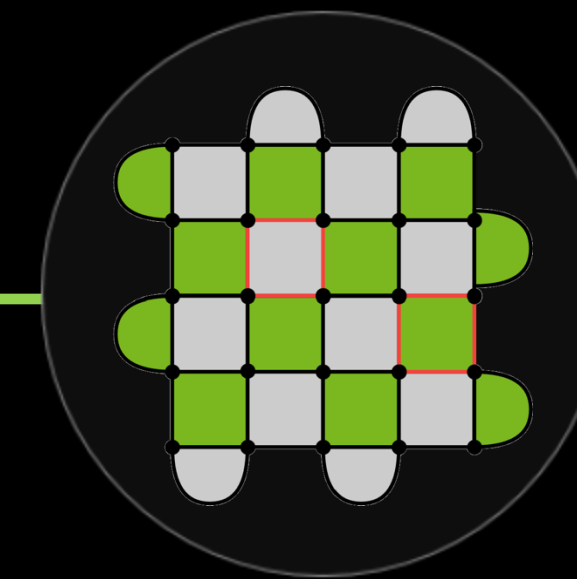
Quantum Algorithm
and Hybrid Application
Development



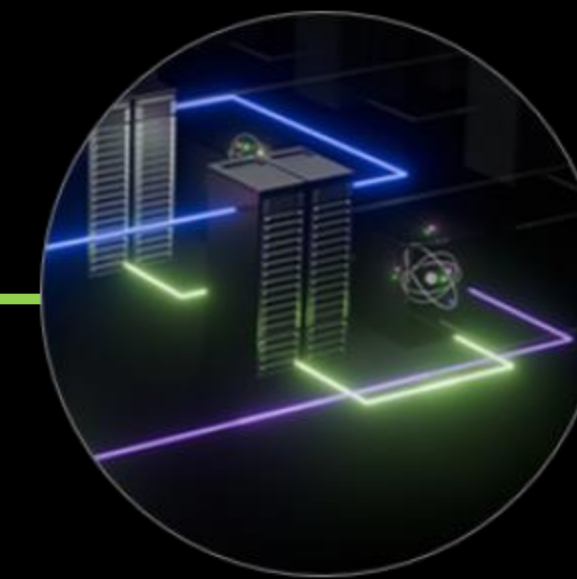
Qubit Design
EDA



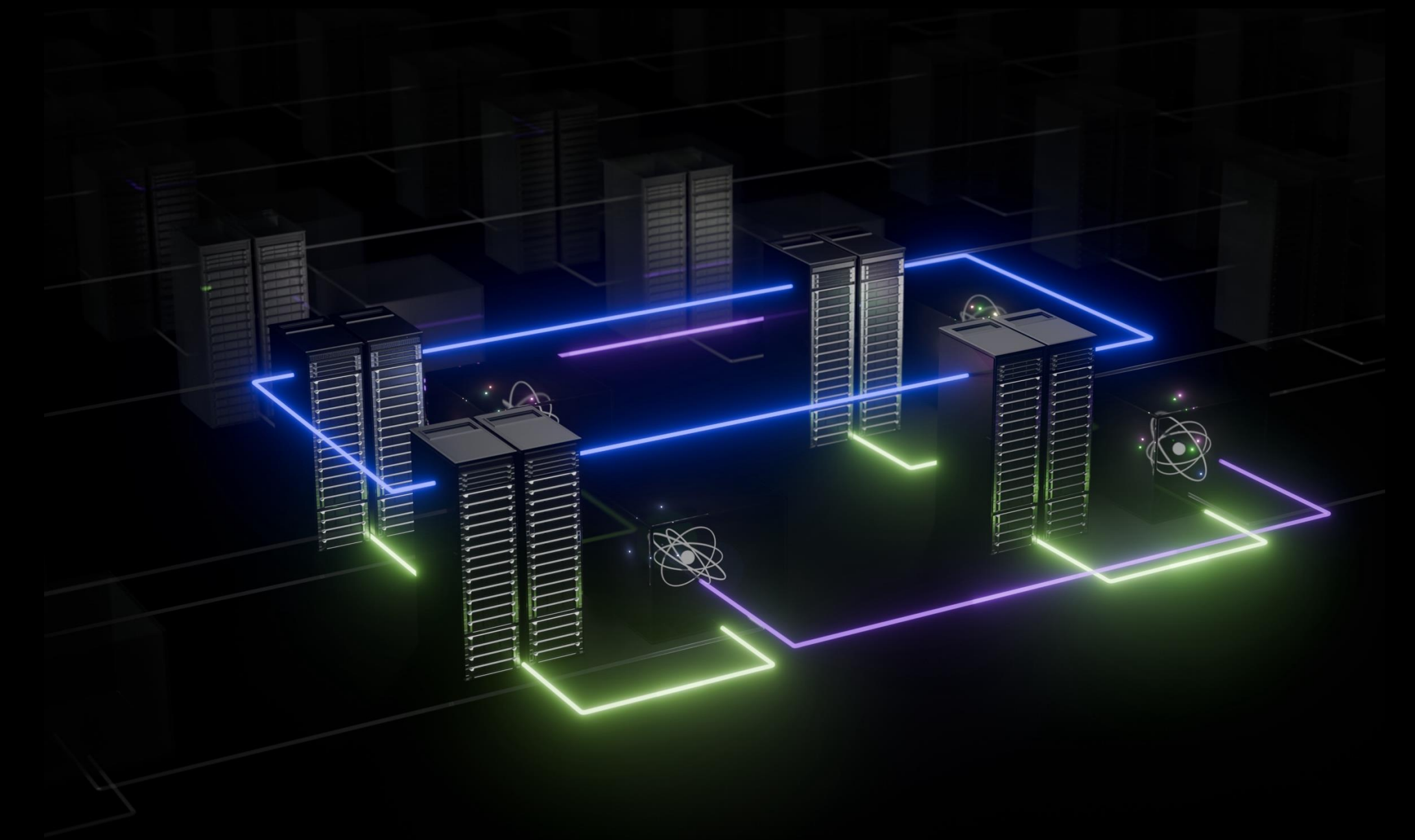
Quantum Data
Generation



Quantum Error
Correction



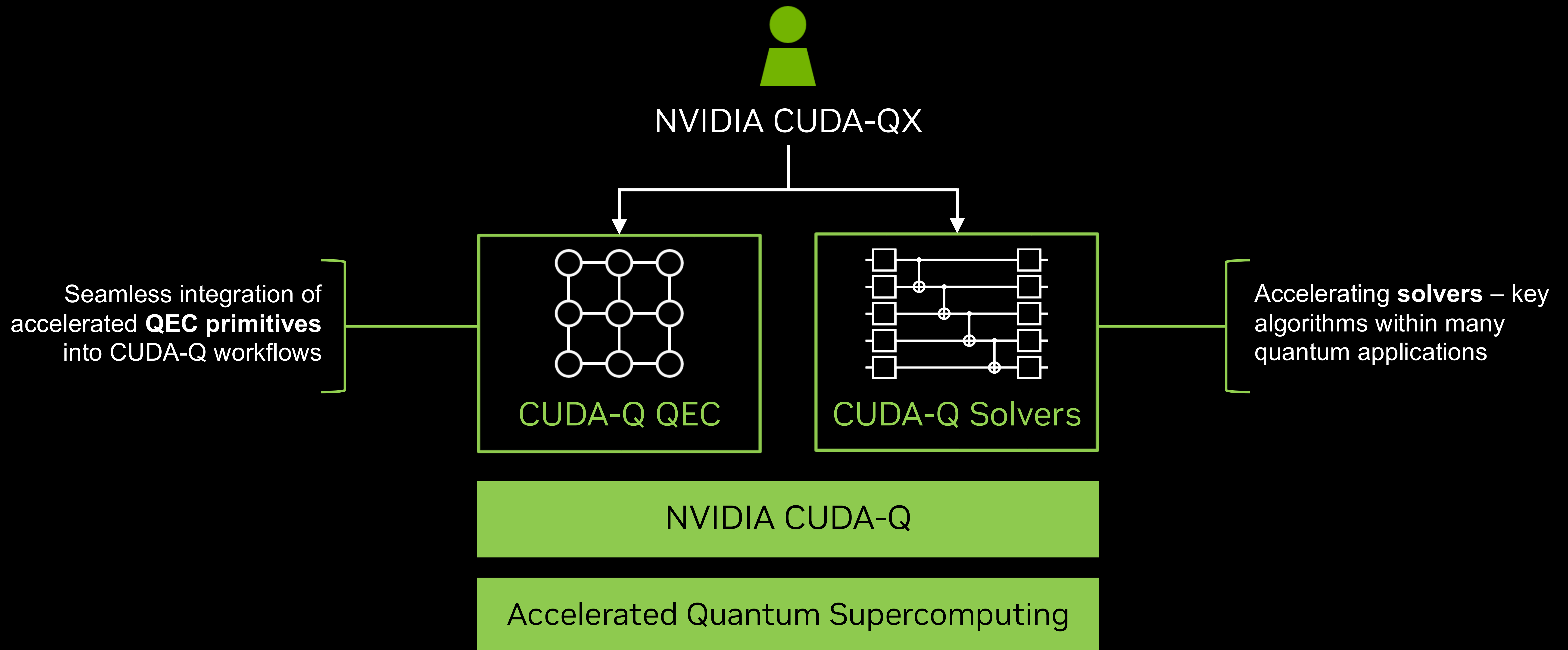
Quantum Computing
Deployments

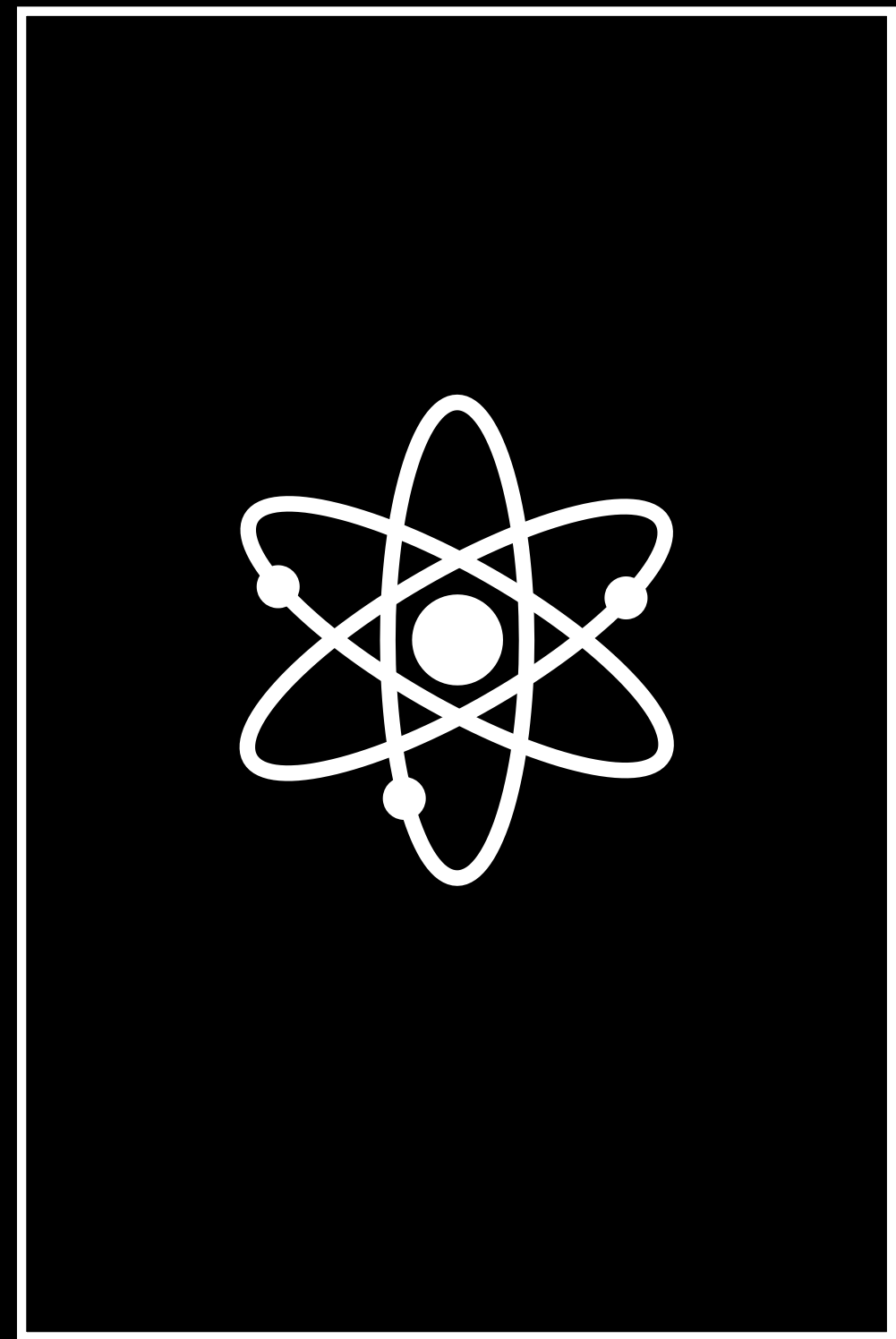


Accelerated Quantum
Supercomputers

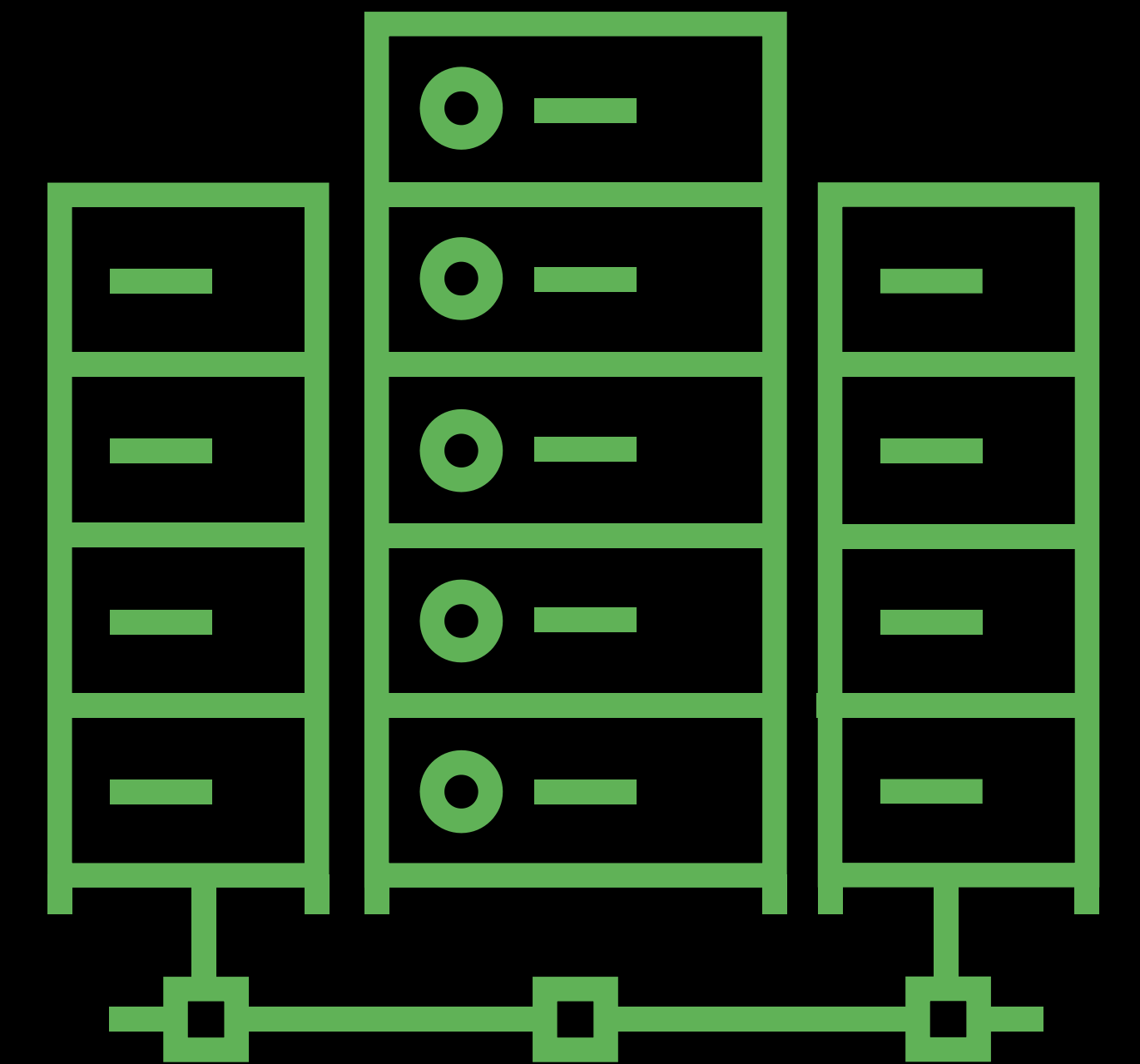
CUDA-Q Libraries

Bringing the power of CUDA-Q directly to quantum computing research problems



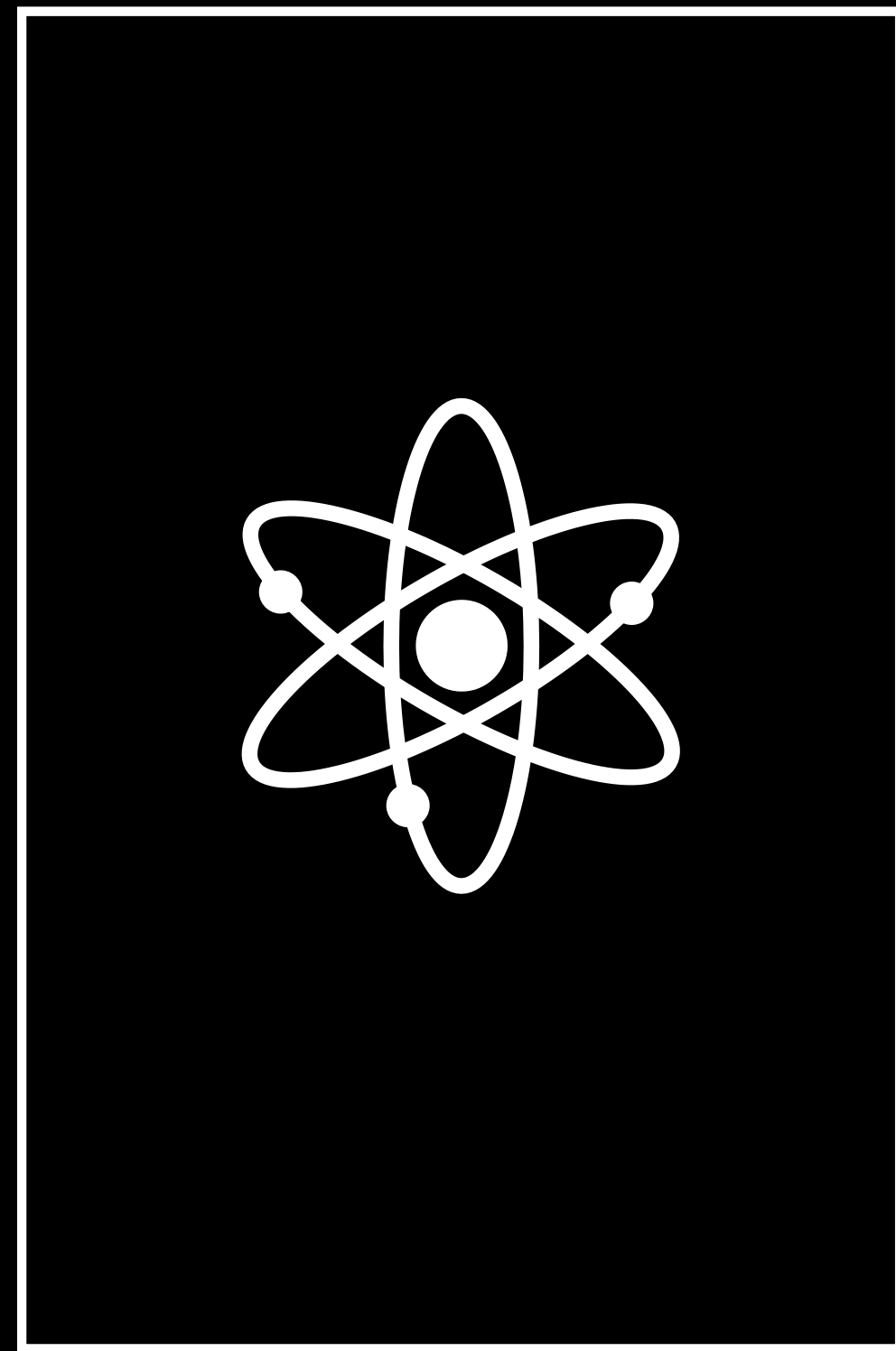


Quantum Processors

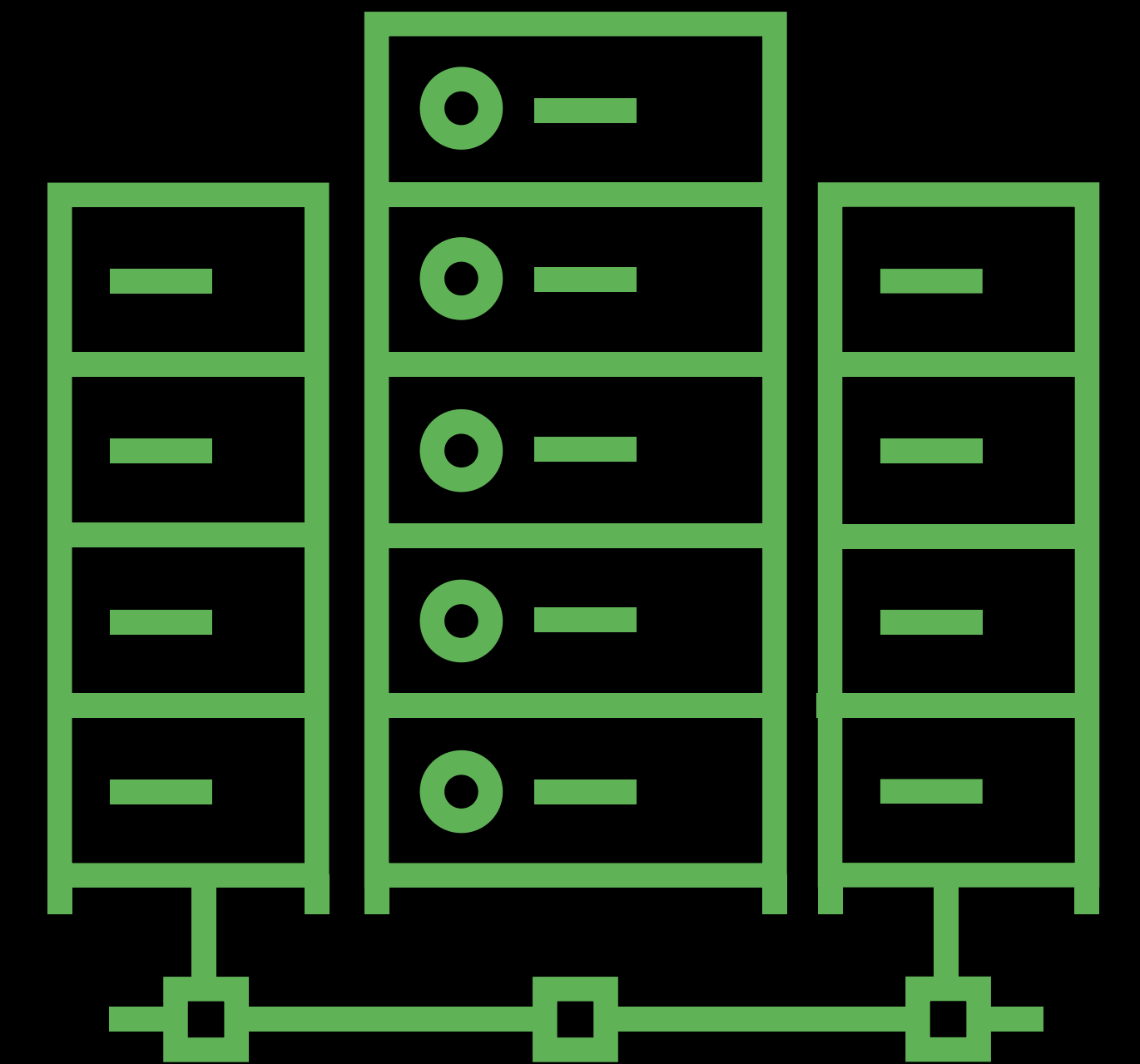
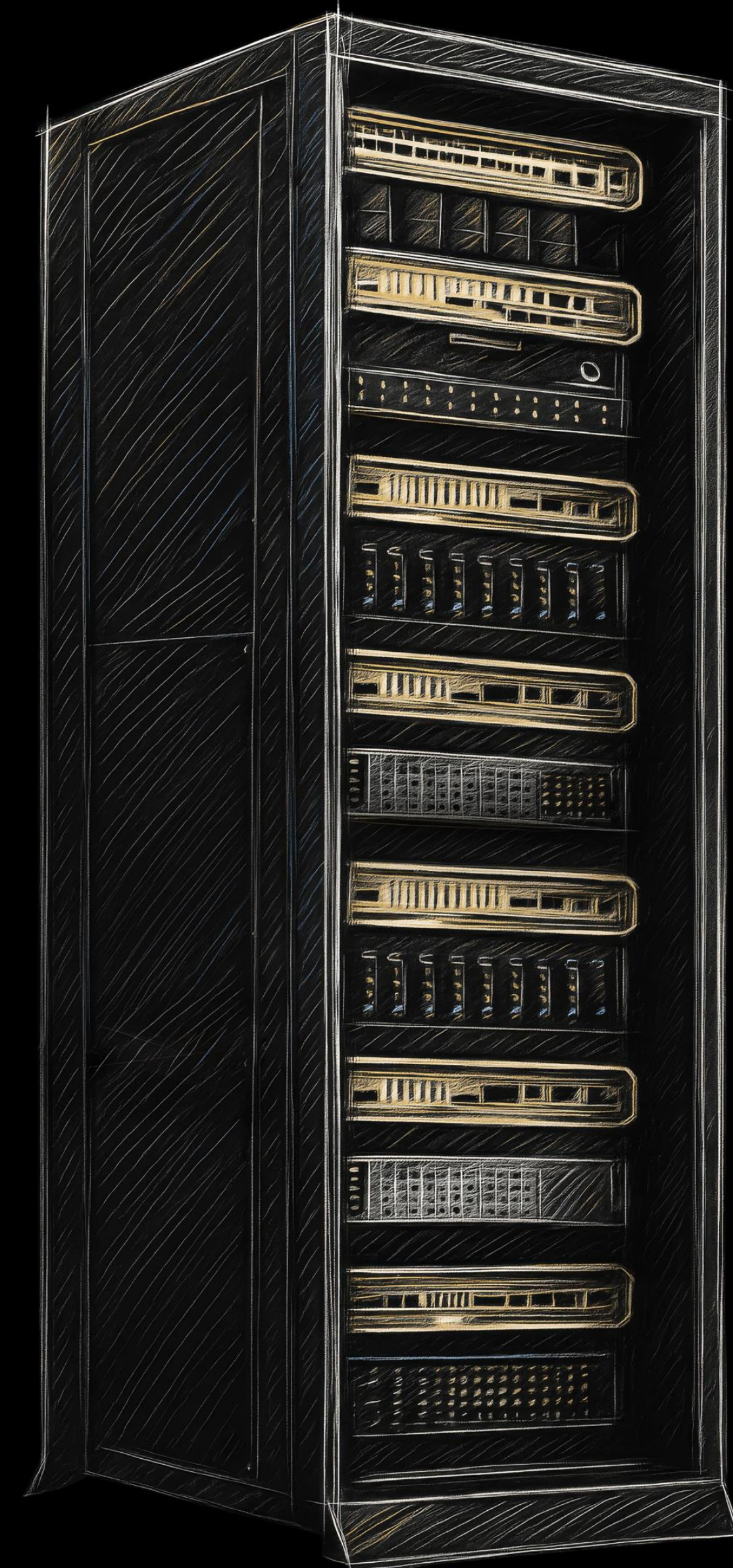


**Supercomputing
Centers**

NVIDIA NVQLink

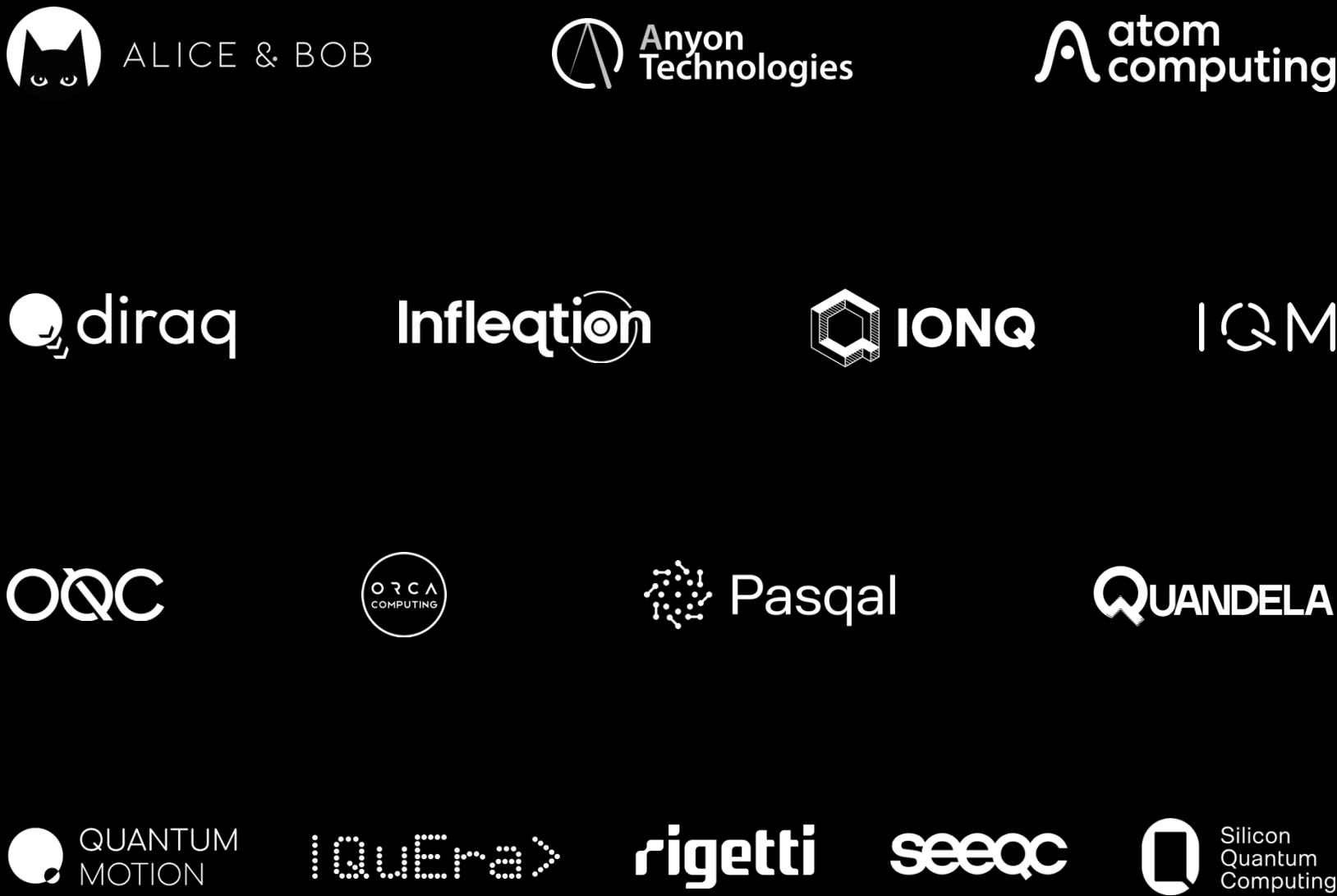


Quantum Processors



**Supercomputing
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Quantum Processors

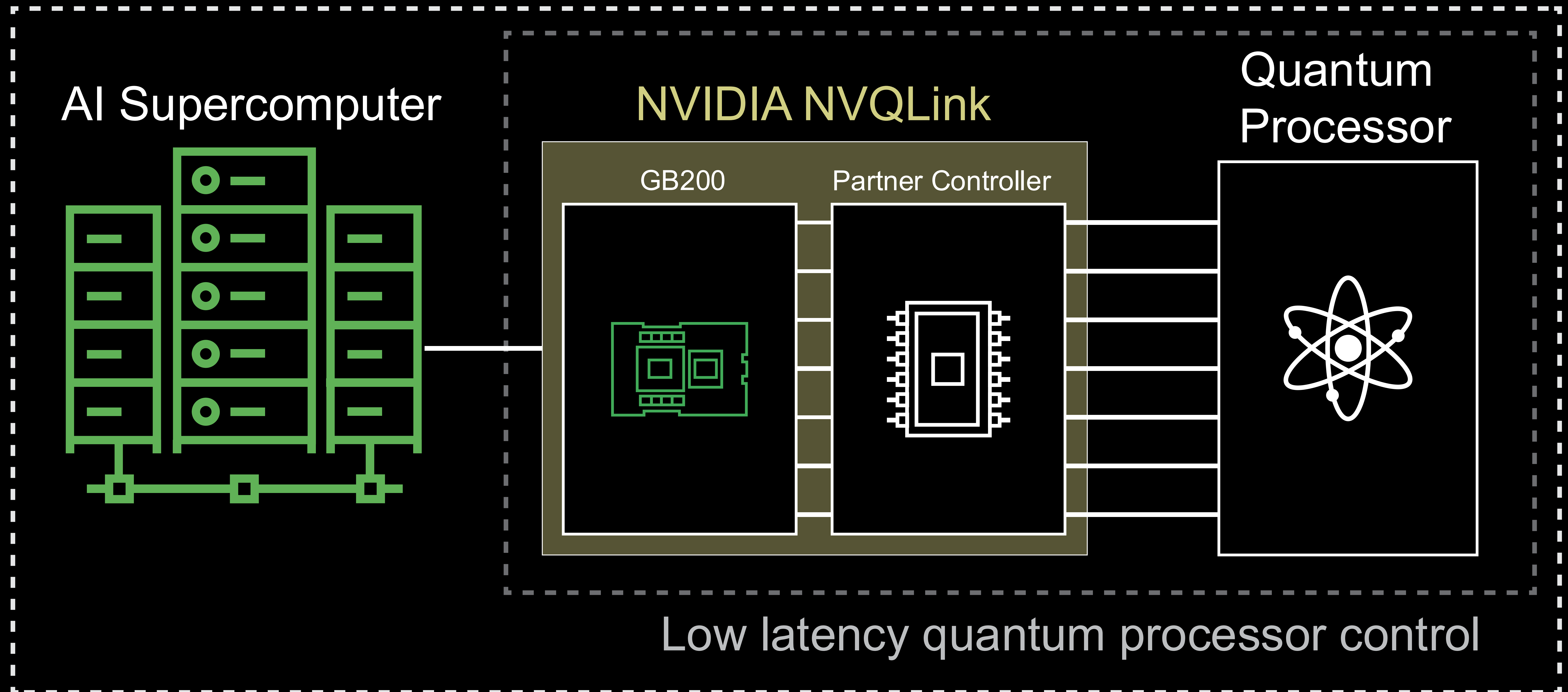


Quantum Controllers



Supercomputing Centers

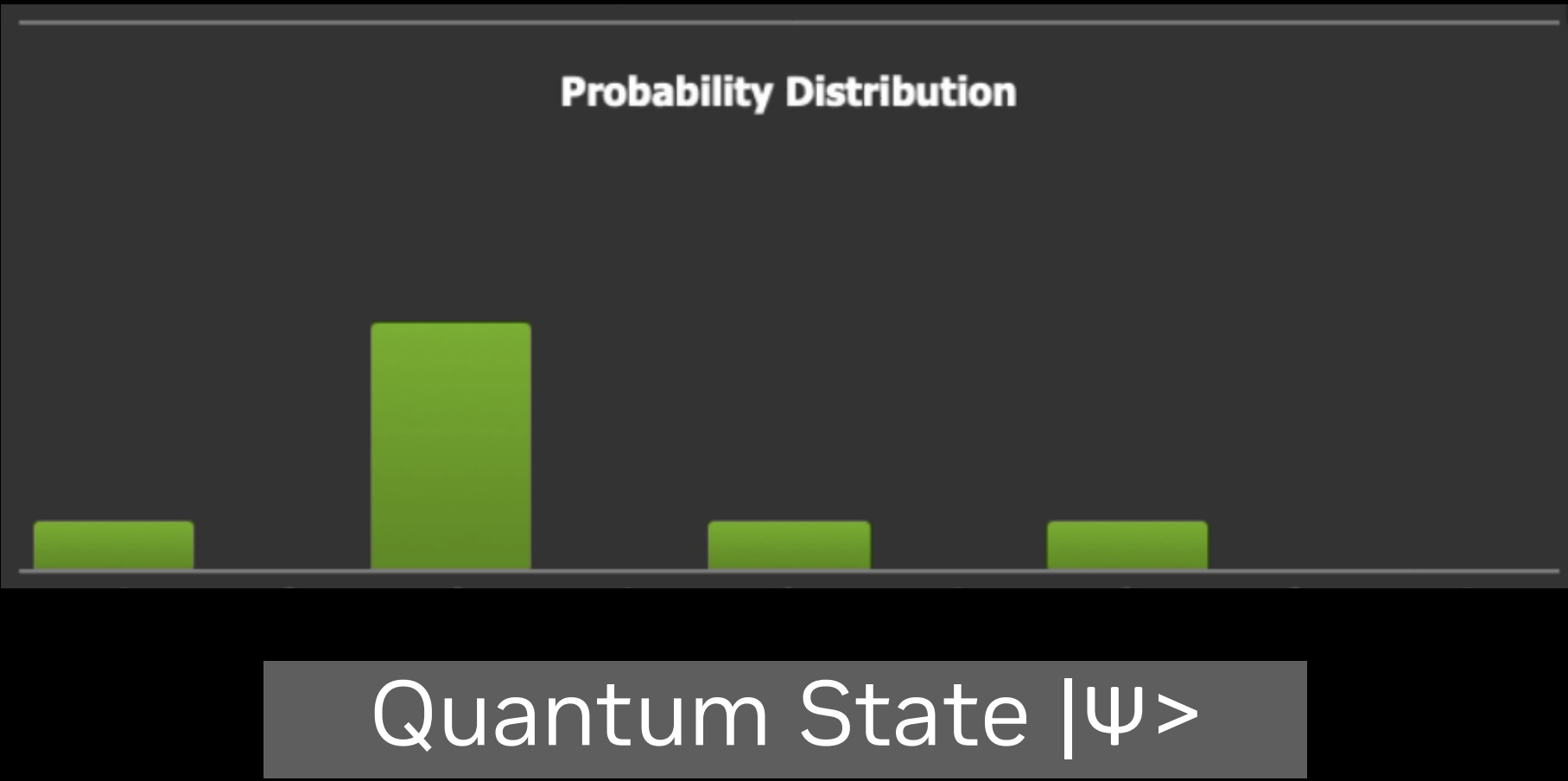
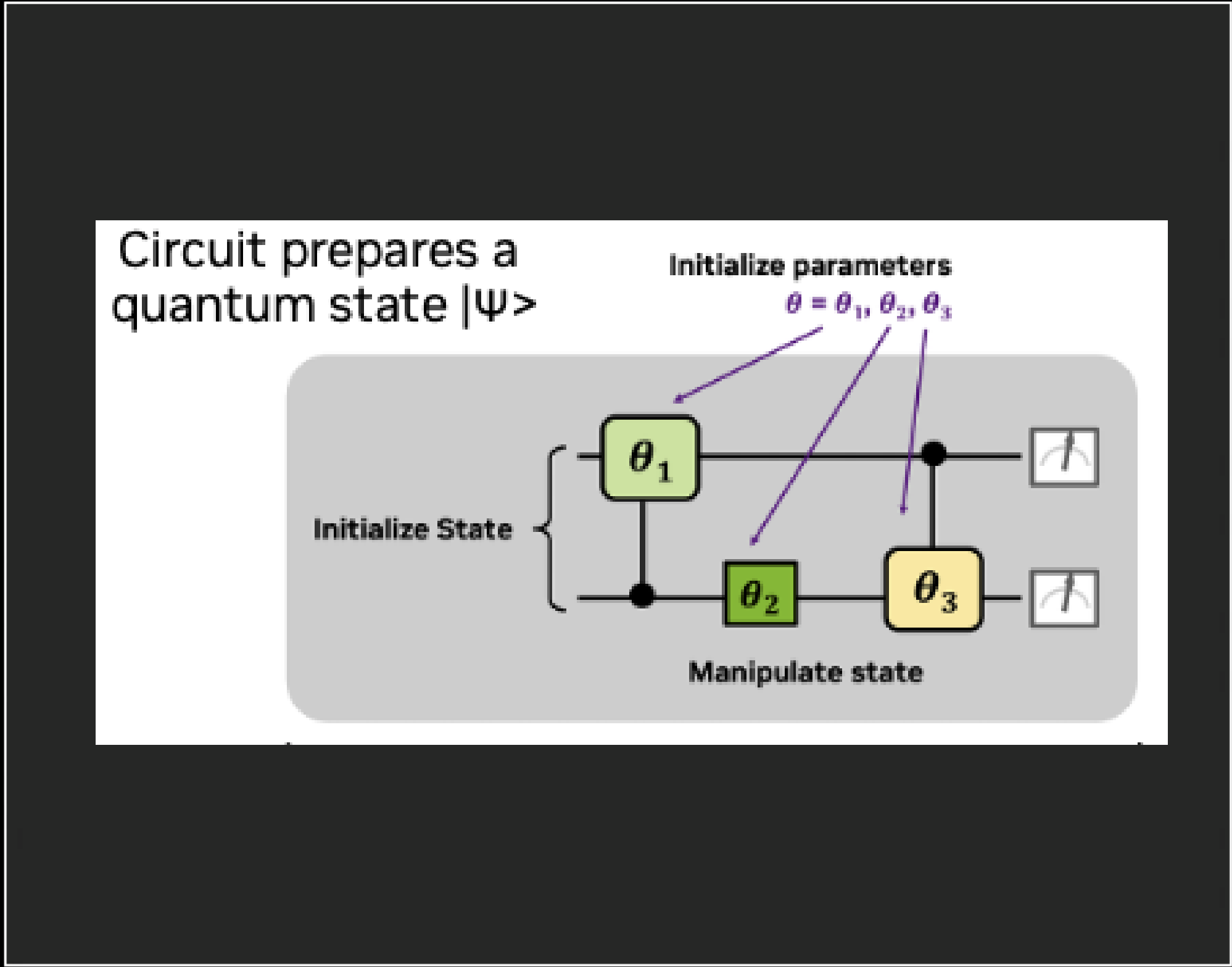
NVIDIA NVQLink



Hybrid quantum-classical computing

Quantum Computation

$$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

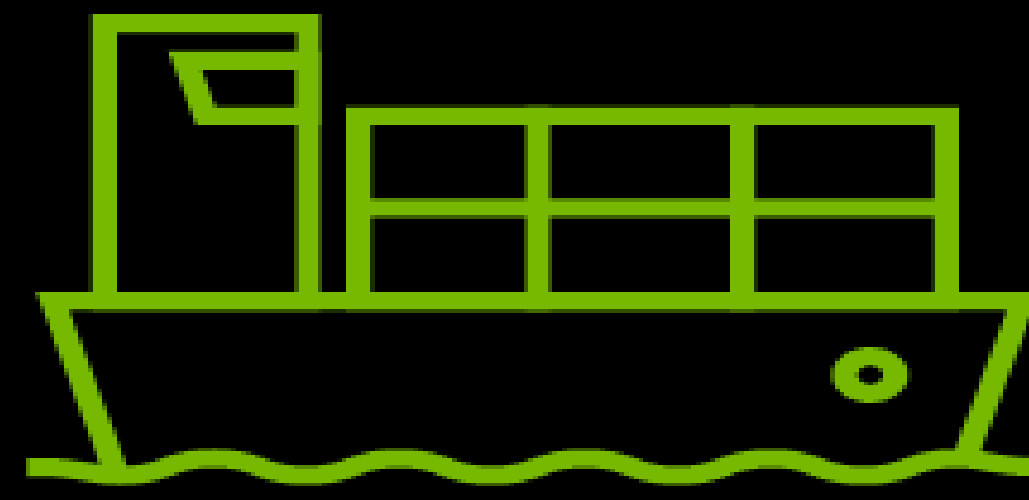


Many Applications – Similar Programming Goals



Finance

Given $|\phi\rangle$, find a circuit that can prepare it. I.e., Minimize difference between $|\psi\rangle$ and $|\phi\rangle$



Logistics

Prepare $|\psi\rangle$ so that when measured the most likely outcome minimizes a cost function

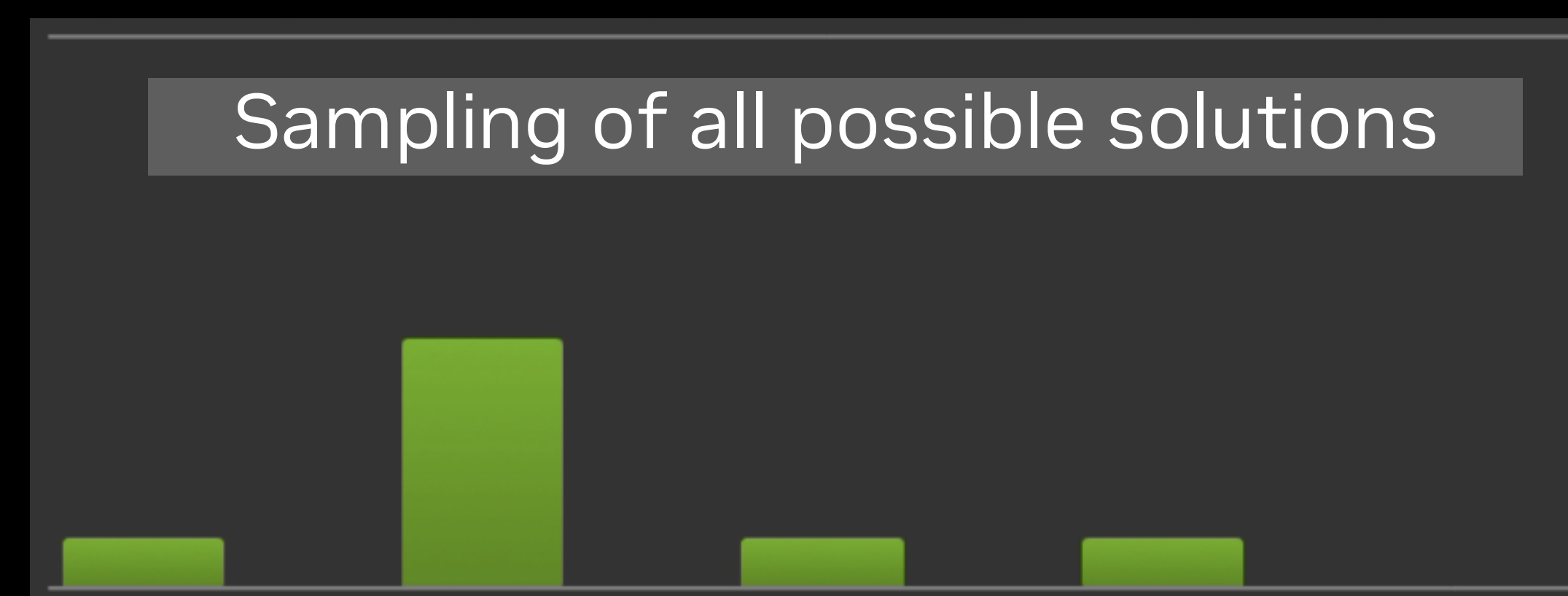
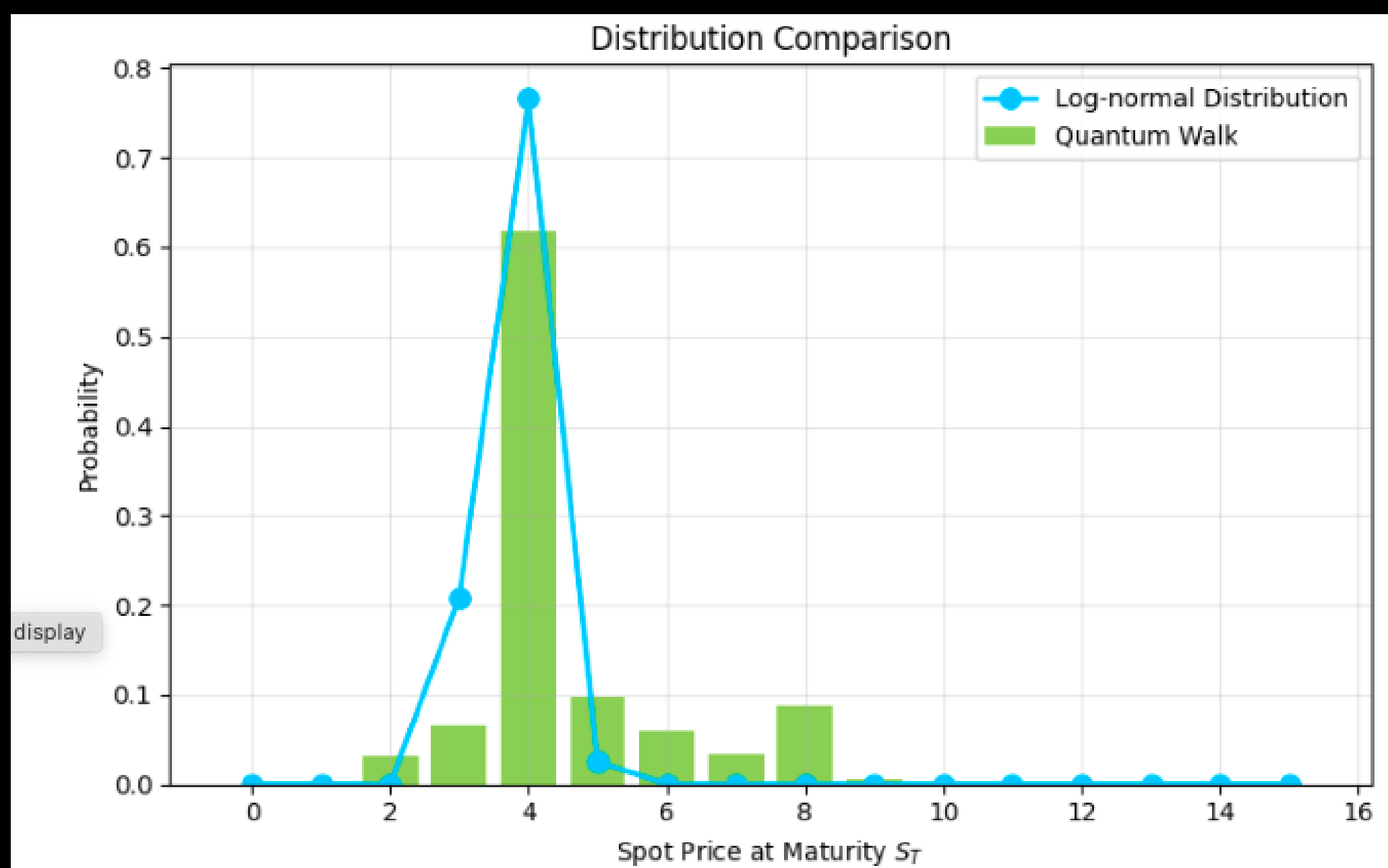


EV
chemistry

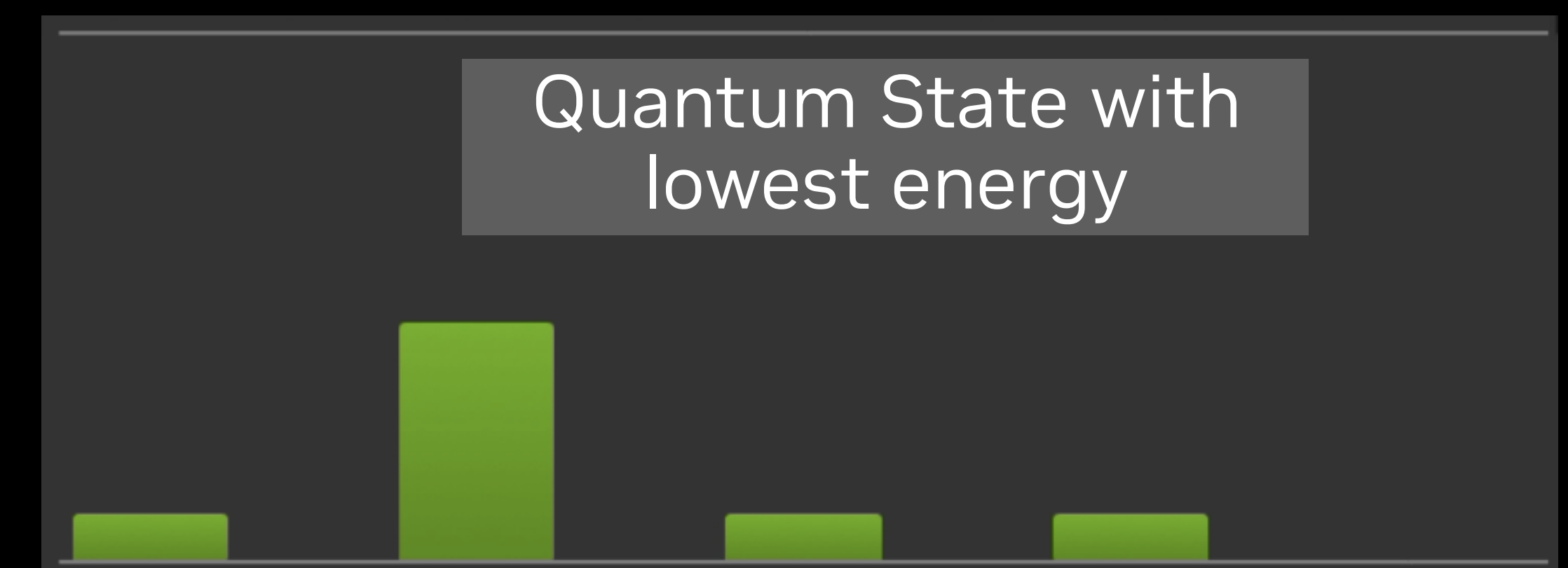
Prepare $|\psi\rangle$, so that the energy of the state is minimized



Drug
discovery



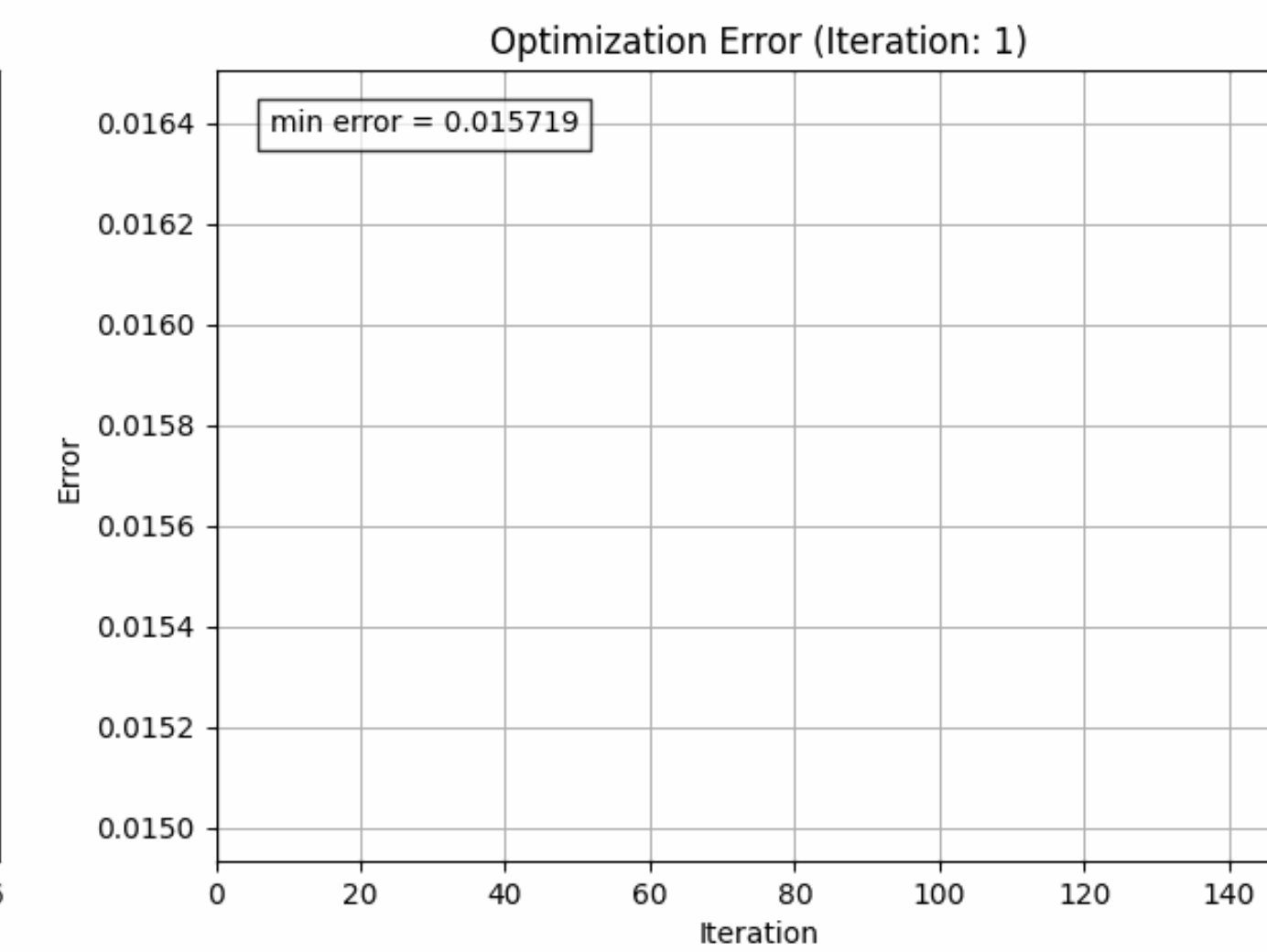
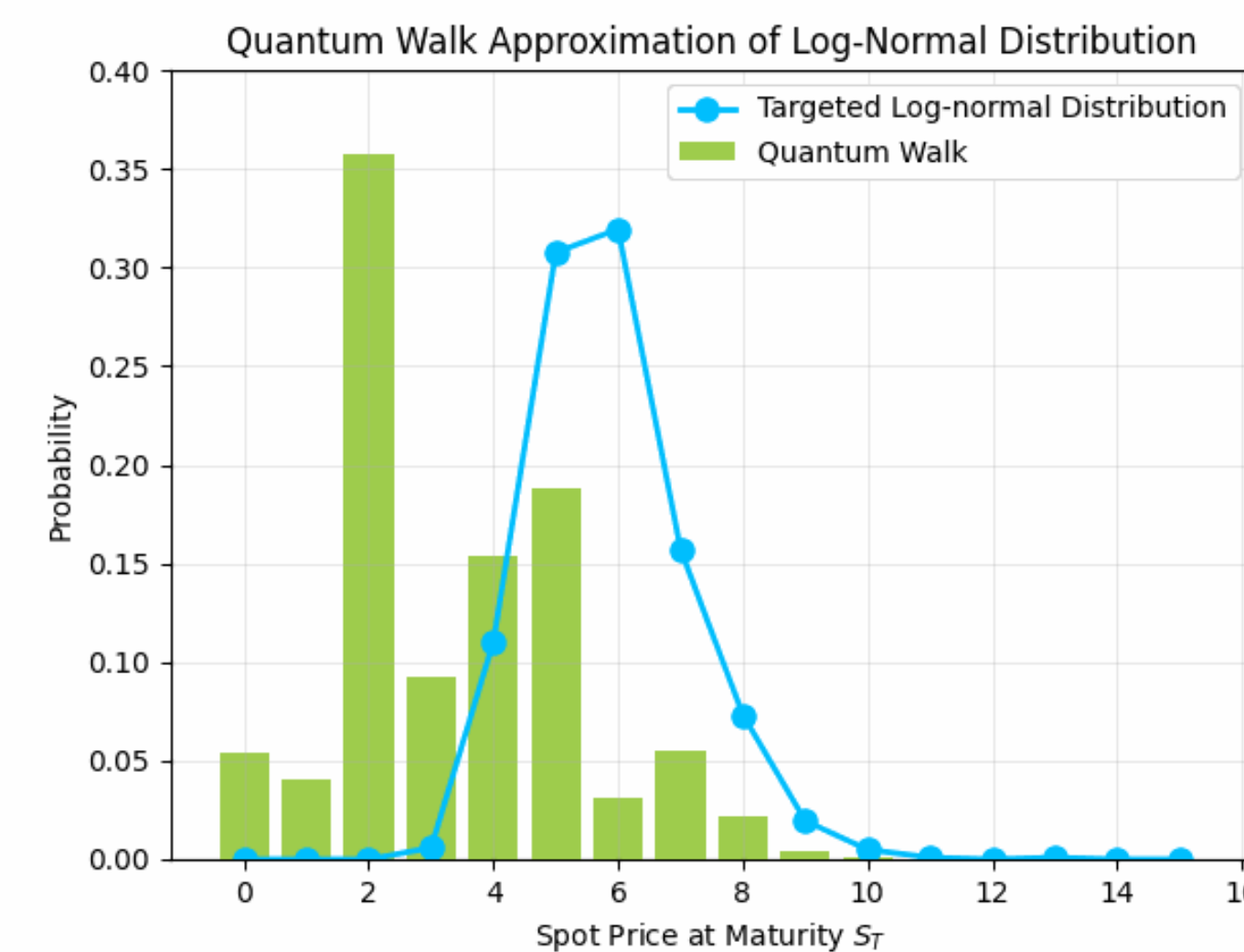
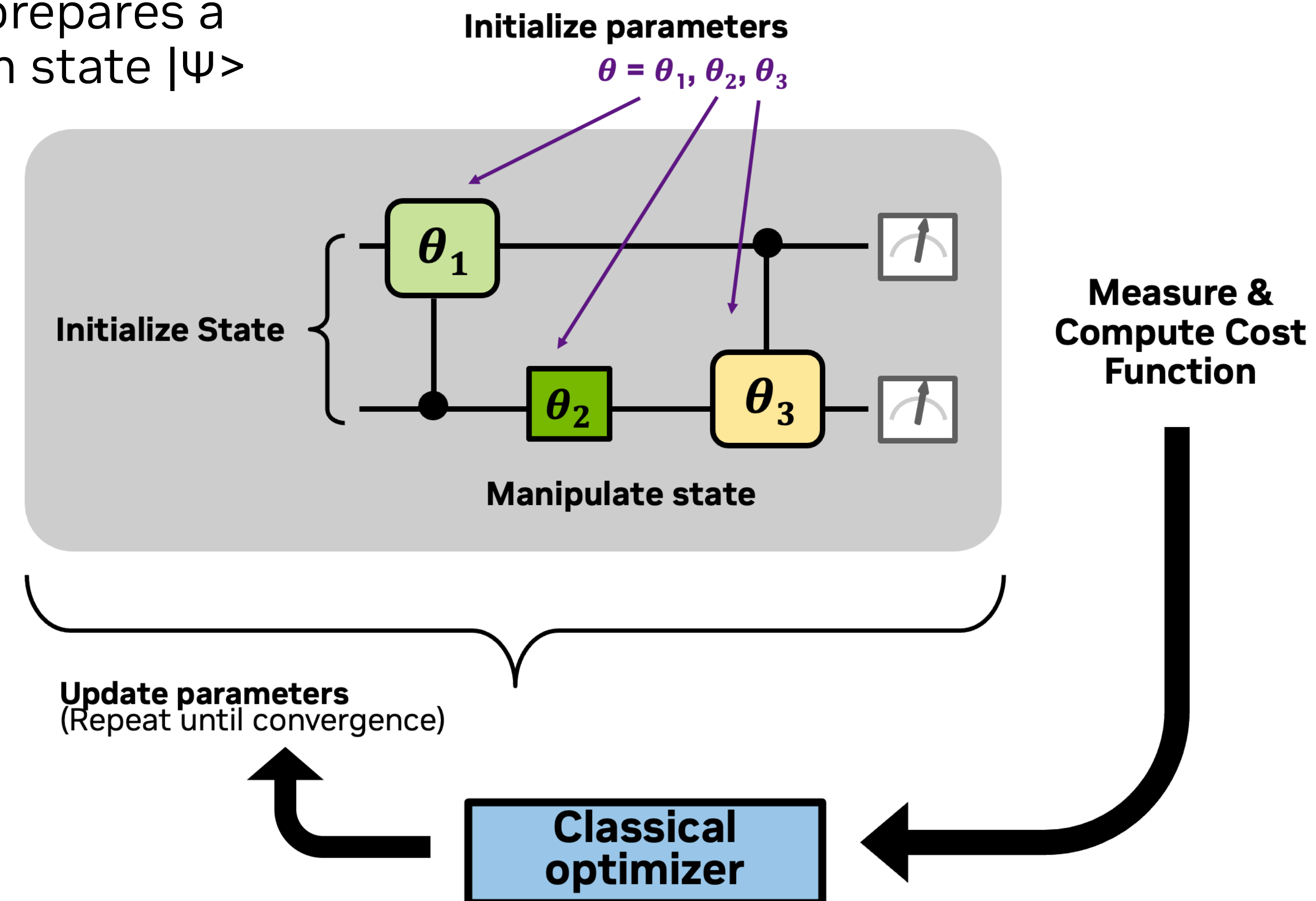
Lowest Cost



Hybrid Quantum-Classical Compute

Variational Quantum Algorithm

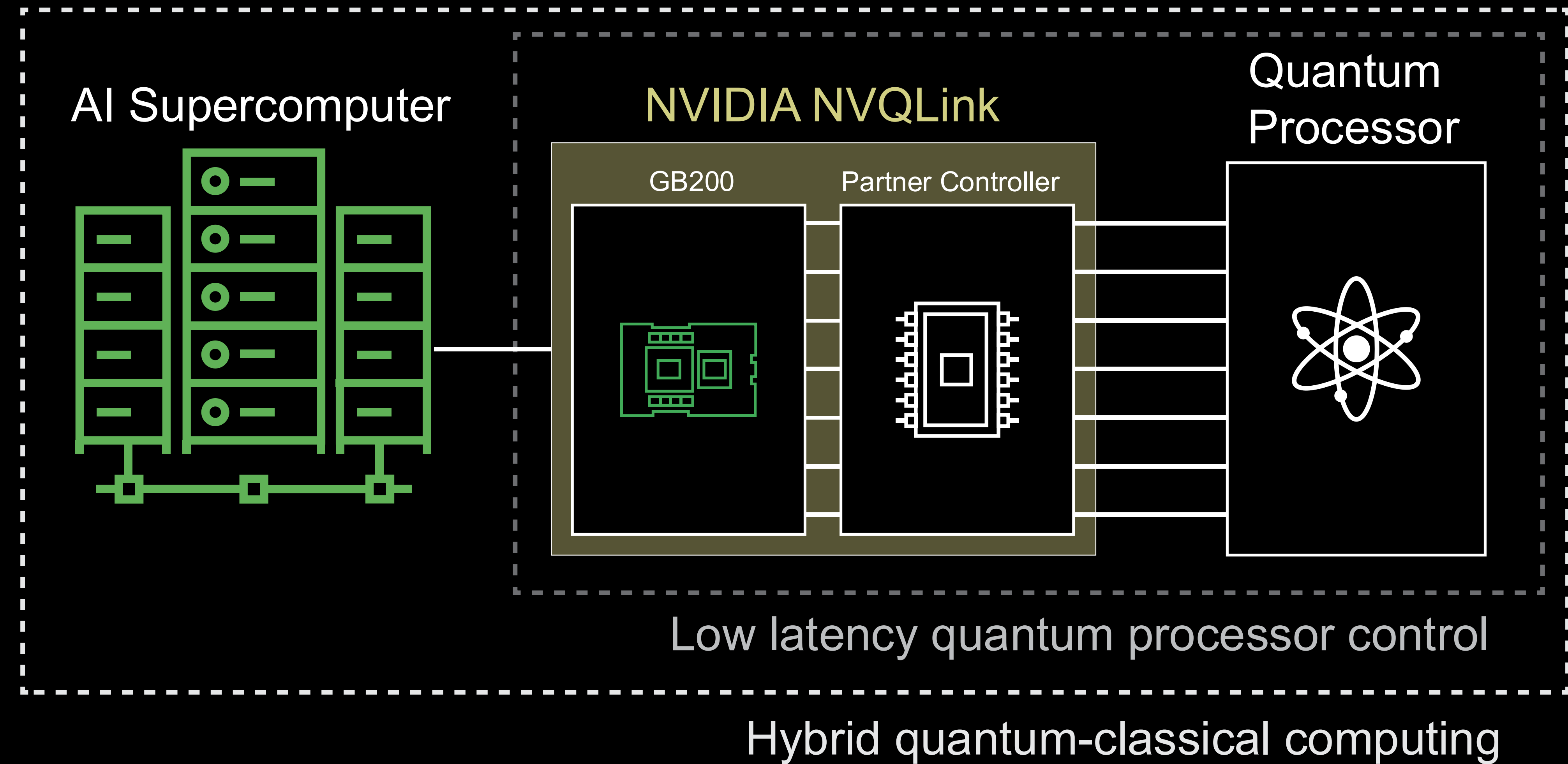
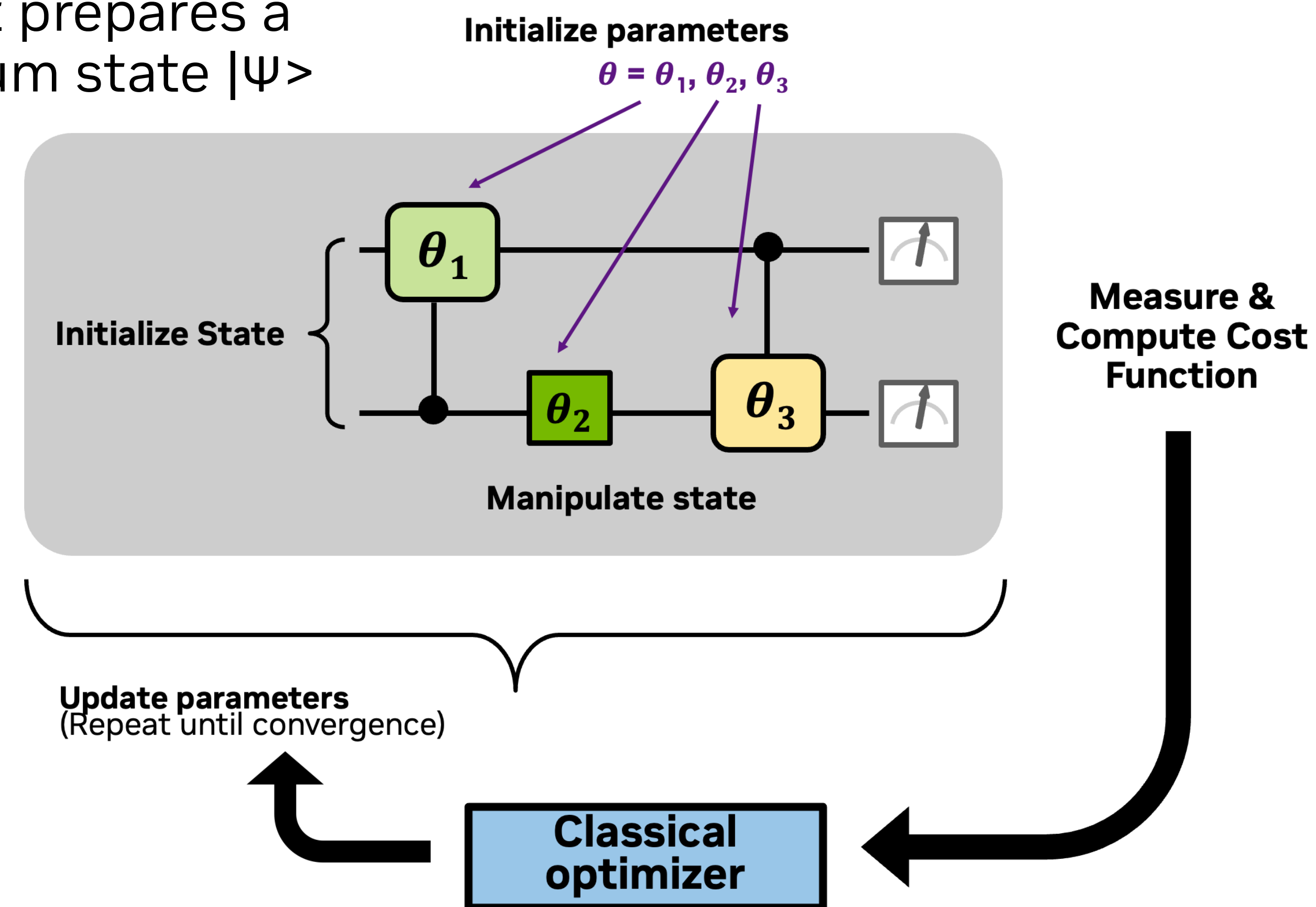
Circuit prepares a quantum state $|\Psi\rangle$



Hybrid Quantum-Classical Compute

Variational Quantum Algorithm

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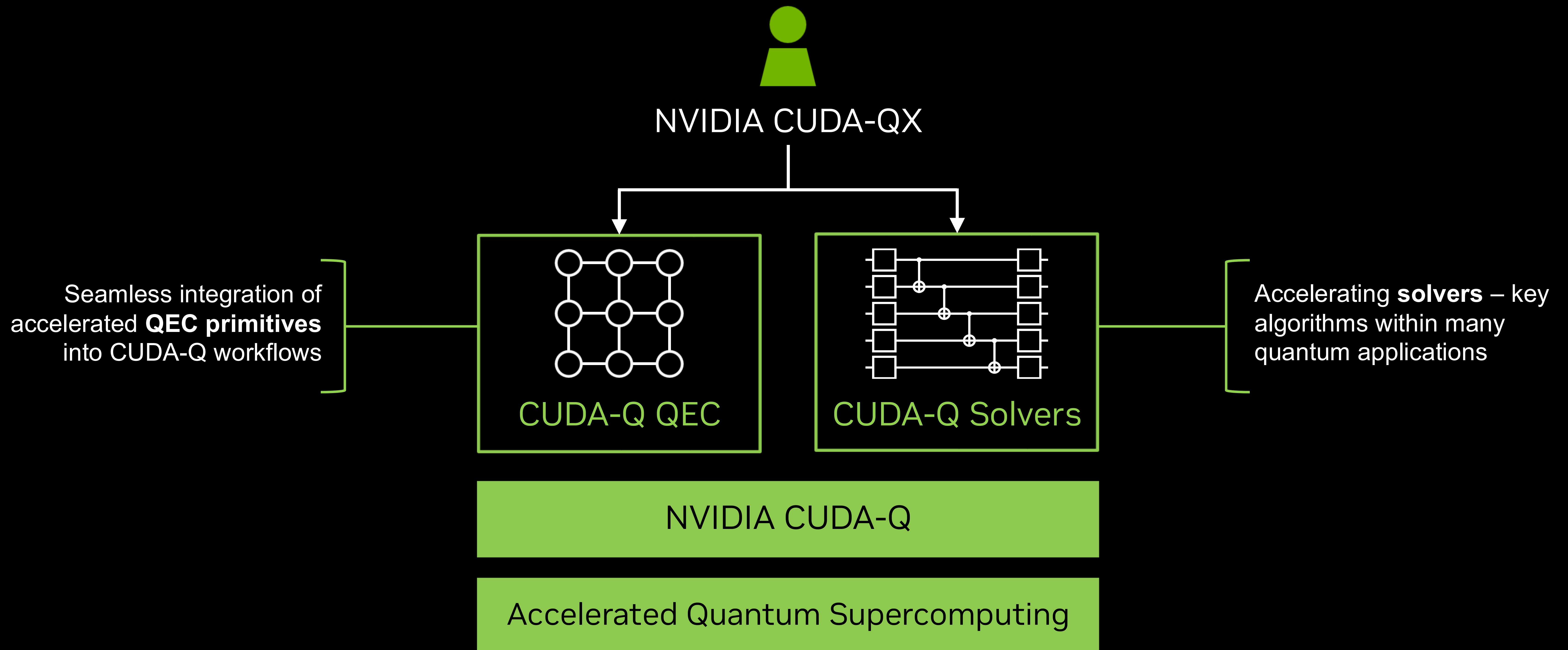


NVIDIA CUDA-Q

Accelerated Quantum Supercomputing

CUDA-Q Libraries

Bringing the power of CUDA-Q directly to quantum computing research problems



Let's Code!

<https://github.com/NERSC/SC25-quantum-tutorial/>



Tutorial Feedback

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- [Resources to learn more](#)

Explore More

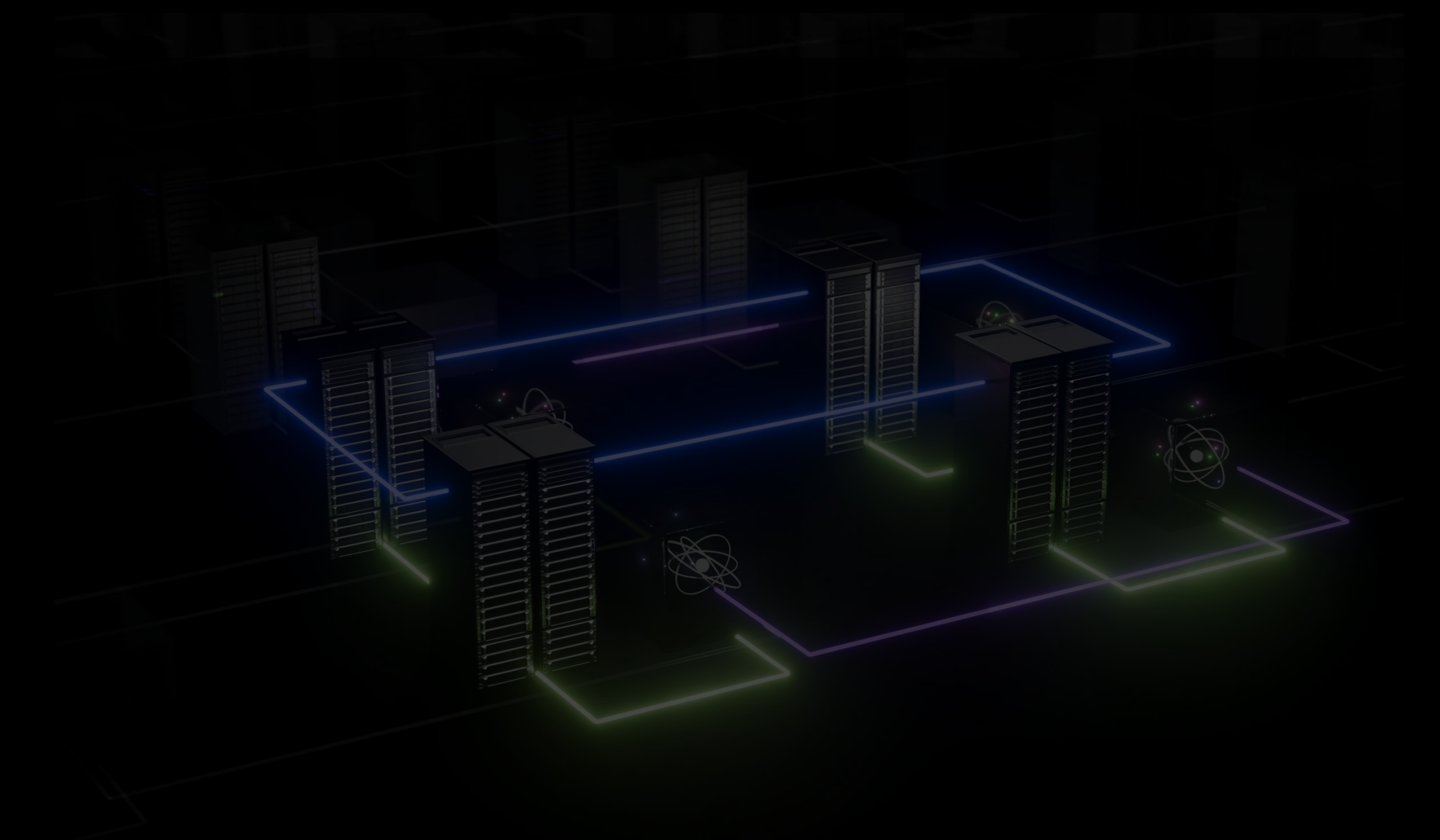
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<http://brev.nvidia.com/>

NVIDIA Launchables Coupon Code
sc-25-quantum

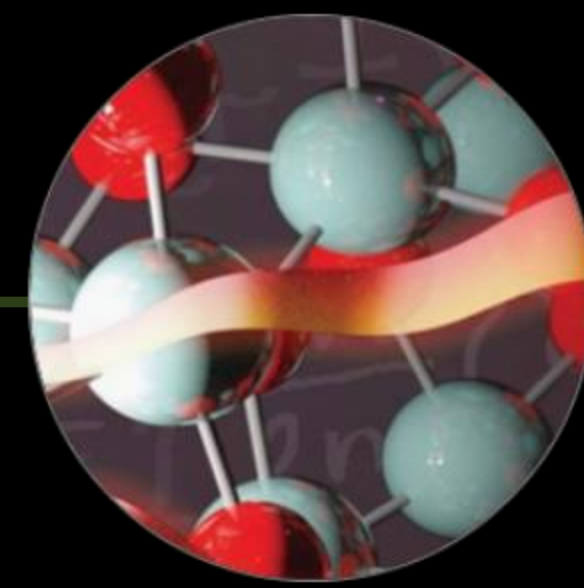
CUDA-Q Academic

Resources for Learning to Program Accelerated Quantum Supercomputers

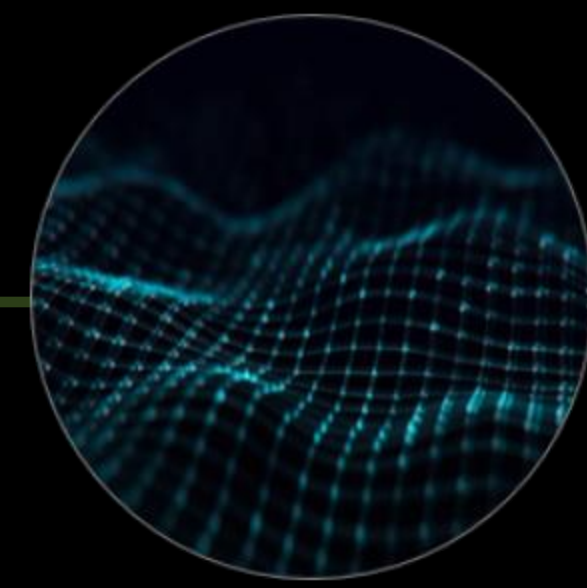


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Accelerated Quantum
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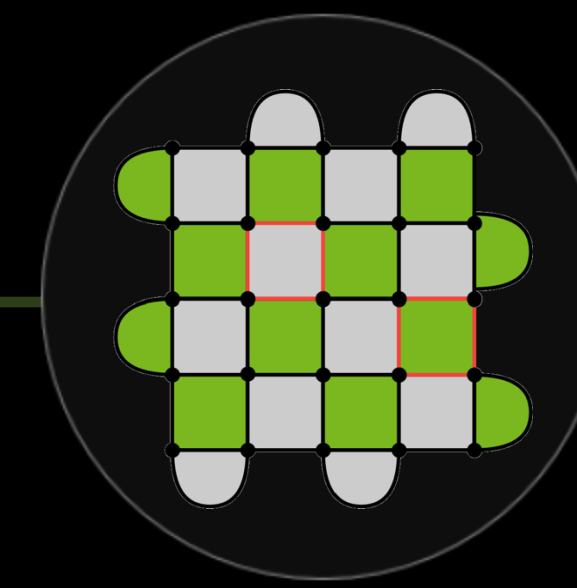
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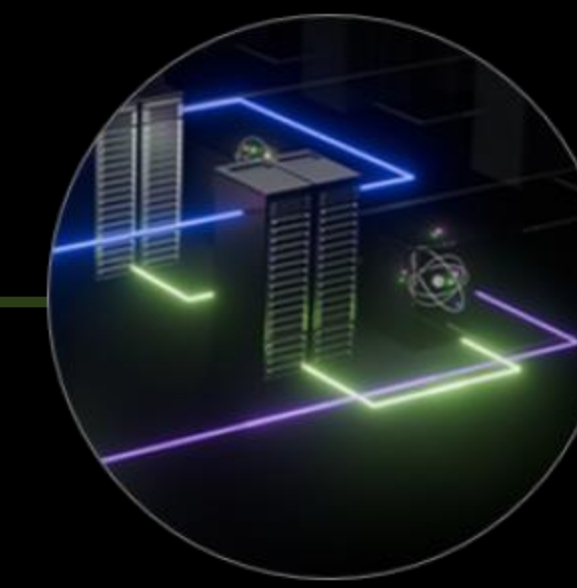
Qubit Design
EDA



GPU Simulations to
Generate
Quantum Data



Quantum Error
Correction



Quantum Computing
Deployments

CUDA-Q
Academic
Modules

Quantum
Applications
to Finance
and
Chemistry

Dynamics 101

Large-Scale
Simulation of
Quantum
Systems

QEC 101

Divide and
Conquer for
Max Cut



Find out more

<https://github.com/NERSC/SC25-quantum-tutorial/>

NVIDIA Quantum

<https://www.nvidia.com/en-us/solutions/quantum-computing/>

CUDA-Q v0.12 Now Available

Python – Install from PyPi

C++ – <https://github.com/NVIDIA/cuda-quantum/releases>

CUDA-Q Academic

Educational resources for CUDA-Q

(<https://github.com/NVIDIA/cuda-q-academic>)



